

MODULE 21

Advanced Queries and Optimization

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

Advanced Queries and Optimization

Query Execution Order

1. FROM / JOIN
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. DISTINCT
7. ORDER BY
8. LIMIT / OFFSET

Optimization Tips

Use specific columns, not SELECT *

Filter early with WHERE before joins

Index JOIN and WHERE columns

Use LIMIT when you do not need all rows

Use EXISTS over IN for large subqueries

Prefer JOINS over correlated subqueries

Common Mistakes

```
-- Wrong: breaks index!
WHERE YEAR(date) = 2024

-- Right: uses index
WHERE date BETWEEN '2024-01-01' AND '2024-12-31'

-- Wrong: leading wildcard, no index
WHERE name LIKE '%Kumar'

-- Right: uses index
WHERE name LIKE 'Kumar%'
```

UNION and UNION ALL

UNION: removes duplicates (slower)

UNION ALL: keeps all rows (faster)

```
SELECT name FROM students
UNION ALL
SELECT name FROM alumni;
```

Rules: same column count and compatible types

EXPLAIN Analysis

EXPLAIN SELECT * FROM students WHERE email = 'a@b.com';

type = ref or const --> good (index used)

type = ALL --> bad (full table scan!)

key --> which index was used

rows --> estimated rows examined